

Towards Continuous Patient Care with Remote Guided VR-Therapy

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ABSTRACT

The biomechanically complex hand is crucial to a person's daily life and enables grasping, manipulating objects, and many more tasks. Impairments have a strong impact on a person's independence, quality of life, and overall well-being. Physiotherapy plays a crucial role after surgical interventions. Given the shortage of personnel and therapy session availability, enabling support and monitoring during the absence of the physiotherapist is a key future direction of medical care. Virtual reality (VR) has been shown to be able to support the rehabilitation of hand motor function. However, an individualized and motivating rehabilitation process is crucial to support the affected person until full recovery of the hand function. To address this, we present a prototype of a VR rehabilitation system that allows for the medical expert to a) control exercise planning and to b) receive a detailed report on the patients success. Our rehabilitation assistant is based on a gamified approach that targets the execution of crucial gestures for exercise.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

Physiotherapy and the accompanying rehabilitation process plays an indispensable role in restoring the functionality of musculoskeletal organs. The human hand, being a complex and crucial tool, enables us to perform numerous essential tasks in our daily lives. These tasks, can be simple as opening a door or grasping for food. The ability to perform these tasks is crucial to our independence and quality of life. While physiotherapy can support patients to keep these abilities, limiting resources and the shortage of personnel can reduce the amount of physiotherapy sessions for an individual. To increase the support for the patient and enable a monitoring of correctly executed exercises, augmented reality (AR) or virtual reality (VR) can improve the rehabilitation process or support the patient with exercises at home. Rehabilitation support with AR or VR can be individualized specifically for each patient and can incorporate gamification aspects (i.e., exergaming) [1, 6] and automated monitoring [4]. This enables a more engaging rehabilitation process

and a better monitoring of this process for the therapist. Towards a holistic solution, the perspectives of rehabilitation specialists and the patient needs must be combined to provide focused, therapeutic exercises and pleasant training. To this end, we contribute both an therapist authoring and exercise-analysing rehabilitation approach for VR-assisted hand rehabilitation.

2 RELATED WORK

To restore full function after surgery or slow down limitations caused by chronic conditions such as rheumatism, the right physiotherapy is crucial. KinoHaptics [4] is a wearable system to address post-surgery rehabilitation. This system provides vibrotactile feedback to avoid injuries during the physiotherapy process. Besides direct feedback on ongoing therapy, physiotherapy can be supported via AR or VR [6]. To track hands in VR, smart gloves, the Leap Motion sensor, or camera data have been used [6]. Alimanova et al. [1] use the Leap Motion sensor to track the hand and proposes a desktop application with several minigames. Pereira et al. [3] introduce StableHandVR, a serious game using Oculus-driven hand tracking. The participants performed gamification tasks by pinching and poking interactions in VR. Other approaches address arm kinematics, e.g., [2]. However, there is a lack of research for eased expert-controlled planning and analysis. Our prototype aims to foster research in this regard.

3 METHOD

Our approach is split into two parts, the first one is the game for the patient. The second one is the therapist application. For the patient game, we use a tabletop-placed Leap Motion sensor for hand tracking together with an HTC Vive head mounted display (HMD), see Fig. 1. Both devices are connected to Unity 2021.3.12. The therapist application is a web-based dashboard¹. It receives a JSON file from the patient application and visualizes the results for each patient in PowerBI.

The VR application offers two games in the main menu, addressing the hand movements of pinching and clenching a fist. Moving the wrist as a pointer operates the user interface to launch one of the two games. The two endpoints of each movement (i.e., pinching and clenching a fist) are calibrated individually for each user's abilities. For the pinching game, the trajectory of a dart changes continuously, see Fig. 1. One can throw the dart at the trajectory by opening the pinch gesture. In the clenching fist game, the user crushes rocks by clenching the fist in a desert scene. To analyze the correct execution of each hand movements (i.e., by playing each games) we use dynamic time warping (DTW) analysis of the positions of finger joints

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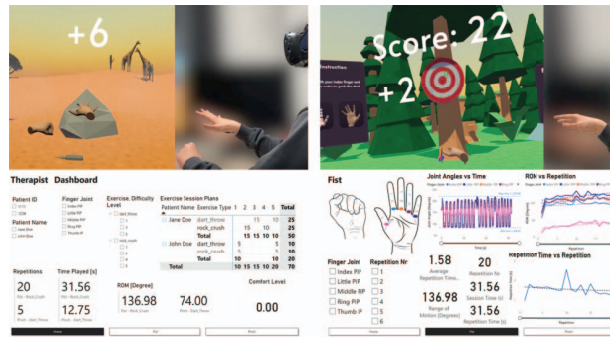


Figure 1: **VR rehabilitation games** (top). When clenching a fist or opening the pinch gesture at the correct moment, patients can crush stones and throw darts to win points in each exercise game. The therapist receives the evaluation of the kinematic data, parameterized patients success, and performance values presented on the **therapist dashboard** (bottom).

and angles. At the end of each game, the user answers a traffic light feedback system indicating how comfortable the exercise was.

The results (feedback and biomechanic statistics) of each run can be interpreted by the therapist's application, see Fig. 1. The biomechanic performance indicators include the range of motion, the session time, the number of repetitions, the repetition time, and the average time of a repetition. Based on this information, the next exercises can be parameterized specific to the patient needs (i.e., difficulty and number of repetitions). The parameters are transferred via another JSON file back to the patient's client application for further exercising. This JSON includes the range of motion (i.e., the difference between the MAX and MIN angle possible by the joint.), the session time, the number of repetitions, repetition time, average for a repetition, joint angles vs time, range of motion vs. repetition and repetition time vs repetition.

4 EVALUATION

4.1 Preliminary User Feedback

To evaluate our approach, we conducted a pilot user study with eight participants (four male and four female). We measured the system usability scale (SUS), the level of virtual embodiment (VE) for the hand model used in the VR game, and collected qualitative feedback. The SUS indicated that both games are have a positive usability as both are rated over the threshold of 68 (Dart game 84.68 ± 9.87 , Stone game 90.00 ± 6.95). Three participants rated the level of virtual embodiment to be little above average (2.58 ± 0.36) according to the hand-related questions of the Virtual Embodiment Questionnaire [5]. Qualitatively three participants struggled with the hand tracking during the dart game. Nevertheless, all participants mentioned, that the games are easy to learn, to use, and are not unnecessarily complex.

4.2 Preliminary Therapist's Feedback

We demonstrated the system to a physiotherapist working in a rehabilitation center. The physiotherapist has hand rehabilitation expertise. In the feedback session, the physiotherapist mentioned that the interface is overall positive, denoting an interest into the possibilities. For future developments, the kinematic analyses should be comparable for multiple exercises and patients. In addition, the biomechanically correct execution should be monitored more precisely (using software and hardware), and supplementary parameters for recording muscle activity could be integrated.

5 DISCUSSION AND FUTURE WORK

During the evaluation the participants mentioned problems with the accuracy of the hand tracking for the pinching game. To address this in the future we plan to use a more accurate tracking, a VR HMD which integrates hand tracking abilities, or both. For the virtual embodiment, the hand was rated average, which shows that the familiarization of the game or the representation of the hand could be improved. In addition to the users, the physiotherapist also showed interest in the application. The physiotherapist pointed out that measuring muscle power could be beneficial to track the patients progress. However, our participants have all been healthy. For future studies, patients as participants are crucial to evaluate the rehabilitation benefit for them. Overall, participants and the physiotherapist rated our approach positively with important aspects for technical improvement and additions towards practical use, which will be part of our future work.

6 CONCLUSION

We proposed a VR-based hand rehabilitation approach considering both, the patient's and the therapist's perspective. Our approach enables patient-specific exercises which can be parameterized and iteratively analyzed to overcome the current personnel limitations in rehabilitation. In our preliminary evaluation, participants liked the games' usability and a physiotherapist gave positive-interested feedback. Moving towards practical application will require more extensive studies with more patients as participants and physiotherapists. Furthermore, we will improve the performance measurements, the approach's usability, and medical significance, possibly also involving other limbs with adapted tracking.

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